Volatility Regimes and Next-Day Market Risk

Independent quantitative research | SPY 2015–2025 | Modular final implementation | 3 November 2026

Abstract

We investigate whether trailing 20-session daily realized volatility is informative about the next trading session's absolute SPY log return. The frozen dataset contains 2,766 adjusted daily prices and yields 2,745 feature–target pairs. Full-sample Pearson correlation is 0.4572; HAC(20)-robust OLS slope is 0.6207 (95% CI [0.415, 0.826]). On a chronological 2023–2025 holdout, OLS achieves the lowest RMSE, while a constant training median achieves the lowest MAE. The forecasting improvement is modest, metric-dependent, and does not establish a profitable trading signal.

1. Research design and data

For adjusted closing price C(t), define log return r(t) = ln[C(t)/C(t−1)]. The feature sigma(t) is the sample standard deviation of the 20 daily returns observed through t; the target is y(t) = |r(t+1)|. Annualization by sqrt(252) is applied only to visualization. The frozen CSV enables offline reproduction. Missing rolling-window and terminal target values are removed.

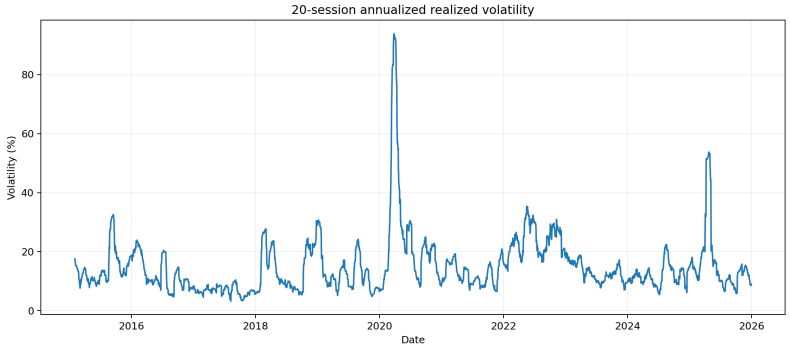


Figure 1. Trailing 20-session daily volatility, annualized for visualization.

2. Descriptive statistical association

Pearson r = 0.4572. Full-sample OLS estimates beta = 0.6207, R-squared = 0.209, and HAC(20) 95% confidence interval [0.415, 0.826]. The confidence interval excludes zero under the HAC inference assumptions. This describes an in-sample association, not a causal effect or an out-of-sample forecasting guarantee.

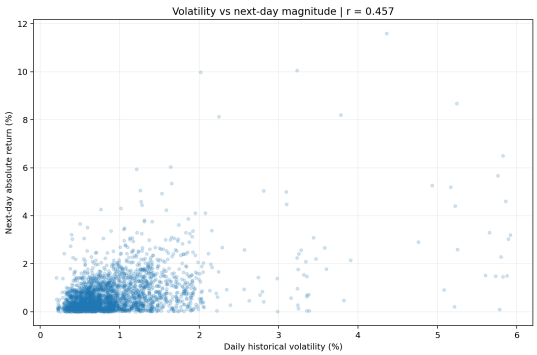


Figure 2. Historical volatility versus next-session absolute return.

3. Chronological holdout and forecasting models

The fixed split trains on 1,993 observations dated 2015–2022 and tests on 751 observations dated 2023–2025. The last pre-2023 feature row is excluded because its target falls in 2023. Model parameters and both constant baselines are estimated on training data only. Model forecasts are clipped at zero to respect nonnegative return magnitude.

Training OLS: predicted y = 0.001191 + 0.659184 × sigma(t). Median quantile regression (q = 0.5): predicted y = 0.000413 + 0.538938 × sigma(t). Baselines use the training-set mean and median of y.

Model	MAE	RMSE
OLS	0.004638	0.006953
Quantile_50	0.004422	0.007036
Mean_baseline	0.004895	0.007109
Median_baseline	0.004406	0.007269

OLS lowers RMSE by 2.20% relative to the training-mean baseline (and MAE by 5.25% against that same baseline). However, the training-median baseline has the best MAE; quantile regression MAE is approximately 0.36% higher than that baseline. Results are fractional log-return errors, not percentages of correct forecasts.

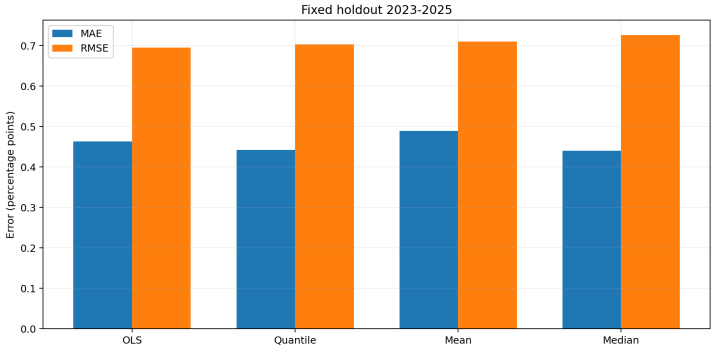


Figure 3. Holdout model comparison; lower MAE and RMSE are better.

4. Temporal and regime diagnostics

Expanding-window annual refits cover 2019–2025. Each model uses only previous years and excludes the boundary feature whose target belongs to the prediction year. The analysis is retrospective, and not an untouched independent confirmation after model exploration.

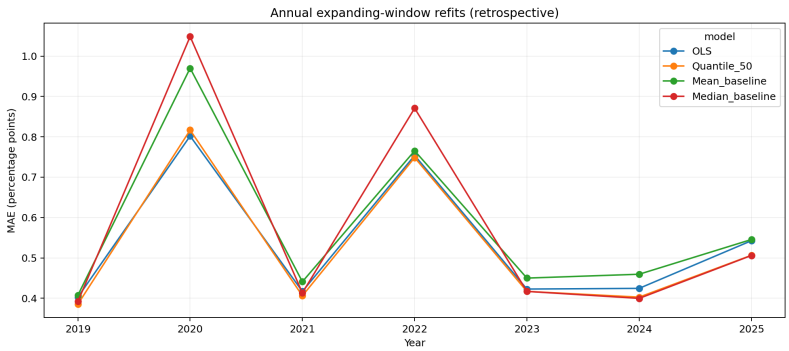


Figure 4. Annual MAE from expanding-window retraining.

5. Volatility-regime analysis

Volatility tercile cutoffs are computed on the training set only: 0.00607 and 0.01082 in daily volatility units. These cutoffs partition holdout observations into low, medium and high historical-volatility regimes. Their mean next-day absolute returns are descriptive, not trading outcomes.

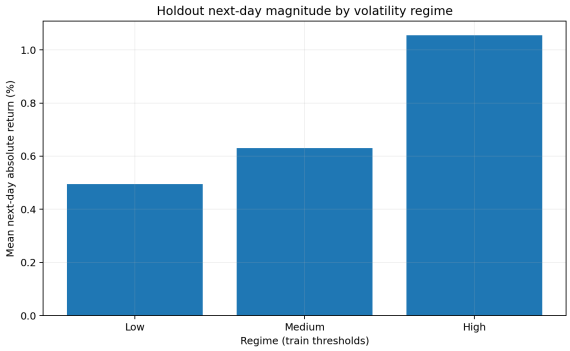


Figure 5. Mean next-day absolute return by training-defined volatility regime.

6. Modular research implementation

The final repository separates the experiment into readable, testable Python modules instead of relying on one monolithic script. The entry point is main.py; run_steps.py prints a guided execution. The former analysis.py remains only as a compatibility wrapper.

Module(s)	Responsibility
config.py, data_loader.py, features.py	Paths/settings; validated CSV loading; return, volatility and target construction
split_data.py, models.py, baselines.py	Leakage-aware train/test split; OLS and median quantile models; constant references
predict.py, metrics.py, holdout.py	Nonnegative forecasts; MAE/RMSE; fixed holdout results
inference.py, walk_forward.py, regimes.py	HAC inference; annual refits; train-defined volatility regimes
plots_history.py, plots_models.py	Nine saved visual diagnostics
export_summary.py, tests/test_project.py	JSON summary and three automated unit tests

7. Reproducibility

```
python3 -m pip install -r requirements.txt

python3 run_steps.py # guided step-by-step run

python3 main.py # full analysis, figures and CSV exports

python3 -m unittest discover -s tests -v
```

Outputs include nine images under figures/ and holdout predictions/metrics, expanding-window predictions/metrics, regime summaries and summary.json under results/. The included frozen CSV avoids external download dependencies. The three unit tests check feature chronology, split separation and error metrics.

8. Limitations and interpretation

The study uses one ETF, one predictor and one forecasting horizon. Market data exhibit heteroskedasticity, serial dependence and extreme outcomes. HAC inference concerns coefficient uncertainty, not statistical significance of out-of-sample loss improvements. The 2023–2025 holdout was examined during development; additional comparisons should be interpreted as exploratory. No direction forecast, execution model, transaction cost, portfolio return or trading strategy was evaluated.

Further work should benchmark against adaptive forecasts (e.g. rolling medians), alternative windows, GARCH or HAR-RV, and genuinely untouched future observations.

Conclusion

Past SPY volatility contains statistically significant information about the next-session absolute return in this sample. Practical forecasting benefits are limited and depend on the loss metric. The final modular implementation makes the study transparent, reproducible and straightforward to audit, without implying tradable alpha.